



AIR UNIT #20 — SAR

ID:

obs-air-sar

Role: Search & Rescue / Survival Extraction

Tier: Advanced Support

Status: Active



PHYSICAL & TECHNICAL

Length: 3.3 m

Width: 2.4 m

Height: 1.2 m

Mass: 30 kg

Power Source: High-capacity rescue battery

Energy Capacity: 3000 Wh

Energy Consumption: 70 Wh/min

Endurance: 30–45 min

Payload Capacity: 3.0 kg (rescue harness + supply bay)

Modules: SAR harness, thermal search suite

Navigation System: GPS + Rescue Pathfinding AI

Structural Integrity: 160 HP

Armor Rating: 0.26

Heat Signature: Medium

Noise Signature: Medium

Top Speed: 24 m/s

Acceleration: Medium

Turning Radius: Tight

Climb Rate: High

Stability Rating: Very High (hover-lock for extraction)



AUTONOMY & AI BEHAVIOR

Behavior Profile: Search & Rescue / Extraction

Aggression: 0.0

Caution: 0.9

Persistence: 0.95

Flanking Likelihood: 0.0

Retreat Threshold: 0.85

Target Priority: Survivors / distress signals

Formation Discipline: Medium

Risk Tolerance: Medium

Sensors

Visual Range: 1600 m

Thermal Range: 1800 m

Acoustic Range: High (survivor detection)

Radar Range: None

LiDAR Range: Short-range

Signal Detection Range: High

Communication

Mesh Range: 6 km

Relay Capacity: 0

Command Link Type: Rescue Grid

Network Priority: High



COMBAT & THREAT MODEL

Firepower Rating: 0

Accuracy: N/A

Stabilization: Very High

Weapon Mounts: None

Weapon Types: None

Suppression Value: 0

Tracking Strength: High

Disruption Strength: Low

Breach Capability: None

Countermeasure Rating: Medium

Threat Modeling

Threat Level: 3

Engagement Distance: N/A

Preferred Terrain: Rescue corridors / disaster zones

Preferred Targets: None

Vulnerability Profile: Ballistic / weather

SQUAD & COMMAND INTEGRATION

Squad Role: Search & Rescue

Squad Size Contribution: 1

Formation Behavior: SAR spacing

Intent Interpretation Rules: Prioritize survivor extraction

Retreat Behavior: Medium

Breach Behavior: None

Flanking Behavior: None

Reinforcement Behavior: Supports rescue, logistics, and command squads

Command Hierarchy

Parent Command Unit: Rescue / Command Core

Subordinate Units: Lifeline, Response

Relay Nodes: None

Network Dependencies: Rescue Grid

VR OPERATOR MODE

Operator Seat Type: Drone View

Operator HUD: SAR Overlay

Camera Feeds: Optical

Thermal View: Yes (high-resolution)

LiDAR View: Yes (short-range)

Sonar View: Yes (acoustic survivor detection)

Tactical Overlay: Survivor markers / extraction zones

Operator Controls

Movement Controls: Full drone flight

Sensor Controls: Thermal / LiDAR / sonar / zoom

Ability Controls: Deploy Harness, Scan, Stabilize

Weapon Controls: None

Stabilization Controls: Auto-hover + extraction lock

Operator Abilities

Survivor Extraction Harness

Thermal Search

Sonar Sweep

LiDAR Scan

Relay Rescue Data

ECONOMY & LOGISTICS

Build Cost: 220 Metal / 230 Electronics

Build Time: 70 sec

Factory Type: Drone Bay

Tech Requirements: Rescue Systems III

Maintenance Cost: Medium

Repair Time: 18 sec

Supply Consumption: Medium

Operational Cost: Moderate

Deployment Requirements: Standard drone pad

Transport Compatibility: All aerial racks

Launch Method: Vertical takeoff

BUILD REQUIREMENTS (PLAYER COLLECTION SYSTEM)

Common Components

16× **Medium Alloy Plates**

12× **Composite Frame Rods**

6× **Micro-Servo Actuators**

3× **Stabilizer Fins**

1× **SAR Harness Housing**

Electronics

16× **Circuit Bundles**

12× **Signal Processors**

4× **Drone Control Chips**

4× **Rescue AI Chips**

Power

1× **High-Capacity Rescue Battery**

3× **Power Regulation Nodes**

Rare / Tradeable

1× **Survivor Detection Core**

1× **Rescue Pathfinding Processor**

Optional Upgrades

Extended Battery Module

Silent Rotor Kit

High-Capacity Harness

Anti-Jamming Shielding